

▼ Figure 54.14 Research Method

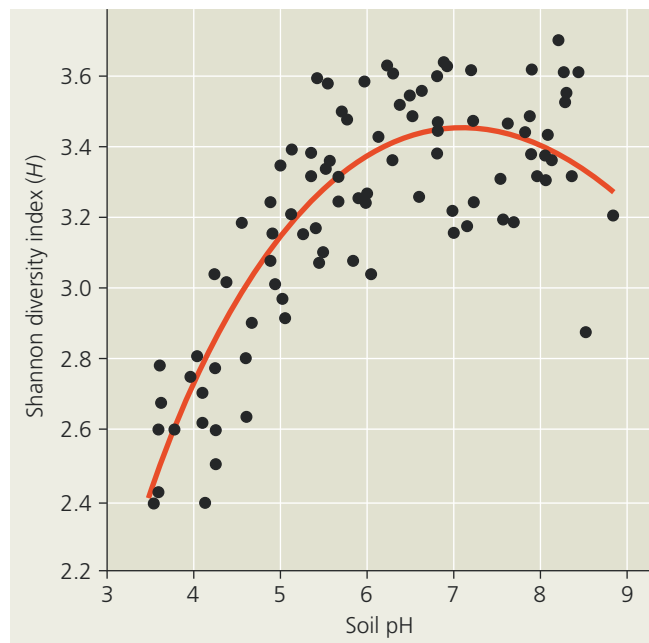
Determining Microbial Diversity Using Molecular Tools

Application Ecologists are increasingly using molecular techniques to determine microbial diversity and richness in environmental samples. One such technique produces a DNA profile for microbial taxa based on sequence variations in the DNA that encodes the small subunit of ribosomal RNA. Researchers used this method to compare the diversity of soil bacteria in 98 habitats across North and South America to help identify environmental variables associated with high bacterial diversity.

Technique Researchers first extracted and purified DNA from the microbial community in each sample. They used the polymerase chain reaction (PCR; see Figure 20.7) to amplify the ribosomal DNA and labeled it with a fluorescent dye.

Restriction enzymes then cut the amplified, labeled DNA into fragments of different lengths, which are separated by gel electrophoresis. (A gel is shown here; see also Figure 20.6.) The number and abundance of these fragments characterize the sample's DNA profile, which was then used to calculate the Shannon diversity index (H) of each sample. Researchers then looked for a correlation between H and several environmental variables, including vegetation type, mean annual temperature and rainfall, and soil acidity.

Results The diversity of bacterial communities was related to soil pH, with the Shannon diversity index being highest in neutral soils (pH 7) and lowest in very acidic soils (pH < 5). Amazonian rain forests, which have extremely high plant and animal diversity, had the most acidic soils and the lowest bacterial diversity of the samples tested.



Data from N. Fierer and R. B. Jackson, The diversity and biogeography of soil bacterial communities, *Proceedings of the National Academy of Sciences USA* 103:626–631 (2006).



► **Figure 54.15** Study plots at the Cedar Creek Ecosystem Science Reserve, site of long-term experiments on manipulating plant diversity.

Diversity and Community Stability

In addition to measuring species diversity, ecologists manipulate diversity in experimental communities in nature and in the laboratory. Many experiments examine the potential benefits of diversity, including increased productivity and stability of biological communities.

Researchers at the Cedar Creek Ecosystem Science Reserve, in Minnesota, have been manipulating plant diversity in experimental communities for more than three decades (**Figure 54.15**). Higher-diversity communities generally are more productive and are better able to withstand and recover from environmental stresses, such as droughts. More diverse communities are also more stable year to year in their productivity. In one decade-long experiment, for instance, researchers at Cedar Creek created 168 plots, each containing 1, 2, 4, 8, or 16 perennial grassland species. The most diverse plots consistently produced more **biomass** (the total mass of all organisms in a habitat) each year than did the single-species plots.

Higher-diversity communities are often more resistant to *introduced species*, which are organisms that humans have moved to regions outside of the species' native range. In research conducted off the coast of Connecticut, scientists created communities of different levels of diversity consisting of sessile marine invertebrates, including tunicates (see Figure 34.5). They then examined how vulnerable these experimental communities were to an introduced tunicate. They found that the introduced tunicate was four times more likely to survive in lower-diversity communities than in higher-diversity ones. The researchers concluded that relatively diverse communities captured more of the resources available in the system, leaving fewer resources for the introduced species and thus decreasing its survival.

Trophic Structure

In addition to species diversity, the structure and dynamics of a community also depend on the feeding relationships between organisms—the **trophic structure** of the community. The transfer of chemical energy from its source in plants and other autotrophs (primary producers) through herbivores (primary consumers) to carnivores (secondary, tertiary, and